

Annie Prasanna Manoharan

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EDUCATION

University of Arkansas, Fayetteville, AR

Aug 2024 – May 2026

Master of Science in Computer Engineering

Coursework: Algorithms, Database Management Systems, Advanced DBMS, Data Mining, Artificial Intelligence, Machine Learning, Image Processing, Cryptography, Privacy Enhancing Techniques

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, C++, C, C#

Software Engineering: Object-Oriented Programming, Data Structures & Algorithms, Software Design, Design Patterns, Multithreading, Concurrency, Exception Handling, Debugging, Troubleshooting, Code Review, SDLC

Data Engineering: ETL, ELT, Data Pipelines, Data Ingestion, Data Extraction, Data Transformation, Data Validation, Data Cleaning, Data Processing, Batch Processing, Real-Time Data Processing, Workflow Automation

Databases: SQL Server, PostgreSQL, MySQL, Oracle, MongoDB, ClickHouse, Relational Databases, NoSQL, Database Design,

Distributed & Streaming Systems: Apache Kafka, Apache Flink, Flink SQL, Event-Driven Architecture, Stream Processing, Event-Time Processing, Windowing, Watermarking, Real-Time Analytics

Workflow & Data Orchestration: Apache Airflow, DAGs, Workflow Orchestration, Scheduling, Monitoring

Methodologies: Agile, Scrum, Software Development Life Cycle (SDLC), Test-Driven Development (TDD), Continuous Integration, Continuous Delivery (CI/CD)

PROJECTS

Real-Time Medical IoT Analytics Platform

Apache Kafka, Apache Flink, SQL, Python, ClickHouse, Grafana

- Developed an end-to-end, sub-second latency streaming platform using Apache Flink SQL and Apache Kafka to ingest, process, and score high-frequency medical IoT device data
- Implemented temporal Flink OVER windows partitioned by device ID to dynamically compute rolling Z-scores over 5-second sliding intervals, applying custom watermark delay strategies to cleanly handle out-of-order data streams
- Designed high-performance ClickHouse schemas using the MergeTree engine sorted by composite keys to physically align disk layouts with clinical querying patterns, while isolating ingestion via an optimized Python multi-topic batch consumer
- Integrated a Grafana visualization layer directly on top of the ClickHouse OLAP tier, delivering real-time clinical dashboards that monitor raw ventilator telemetry and instantly surface streaming Z-score anomalies

Expense Tracker with Predictive Analytics

C++, MySQL, Python, SQL, Database Integration

- Designed and implemented a cross-language expense management application integrating C++ business logic with a MySQL database for storing, retrieving, categorizing, and processing historical expense data.
- Developed and optimized SQL queries to generate category-based expense summaries and reports, integrating database operations with the application's business logic.
- Integrated Python-based analytics with the application workflow to analyze historical spending patterns and generate next-month expense predictions.

PROFESSIONAL EXPERIENCE

Teaching Assistant, Department of Computer Science and Engineering, University of Arkansas

Sep 2024 – May 2026

- Supported 100+ students in Programming Foundations through laboratory instruction, debugging assistance, and mentorship in C++ programming fundamentals.
- Assisted students with control flow, functions, arrays, programming logic, debugging, and problem-solving techniques.
- Supported Database Management Systems instruction for approximately 180 students, assisting with database concepts, assignments, assessments, and technical questions.

Project Engineer, Soliton Technologies, India

July 2022 – June 2024

- Designed and developed a C# WPF desktop application integrating C#, Python, and LabVIEW-based test workflows to streamline application usage and improve test automation.
- Implemented reusable software components and automated workflows within an object-oriented application architecture, improving development and testing efficiency.
- Developed customized software solutions for Medtronic as the sole developer, automating approximately 94% of manual test cases while meeting medical-device quality and validation requirements.
- Integrated automated testing and validation workflows to improve test coverage, reduce manual execution, and support repeatable software verification.
- Collaborated with cross-functional engineering teams to debug system-level issues, troubleshoot application behavior, optimize workflows, and deliver production-ready software.